Interactive To-Do List App

A beginner-friendly JavaScript project using HTML, CSS, and JavaScript

Project Description

This project is a simple task manager that lets users add tasks, mark tasks as completed, delete individual tasks, and clear completed tasks. It uses JavaScript events, arrays, functions, and DOM manipulation to update the page dynamically.

Main Features

- Add a task using the input field
- Mark tasks complete using checkboxes
- Delete individual tasks
- Remove all completed tasks with one click
- Show remaining task count and an empty-state message
- Responsive layout for mobile and desktop

How to Run

1. Create a folder named **todo-list**.
2. Save the following code sections as **index.html**, **style.css**, and **script.js**.
3. Keep all three files in the same folder.
4. Open **index.html** in a browser.

File 1: index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>To-Do List App</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <main class="app">
    <p class="eyebrow">STAY ORGANIZED</p>
    <h1>My To-Do List</h1>
    <p class="subtitle">Small steps make big progress.</p>

    <form id="task-form">
      <input id="task-input" type="text" placeholder="What do you need to do?"
             aria-label="New task" maxlength="120" required>
      <button type="submit">Add Task</button>
    </form>

    <div class="summary">
      <span id="task-count">0 tasks</span>
      <button id="clear-completed" class="text-button" type="button">
        Clear completed
      </button>
    </div>

    <ul id="task-list" aria-live="polite"></ul>
    <p id="empty-message">No tasks yet. Add your first task above!</p>
  </main>
  <script src="script.js"></script>
</body>
</html>
```

File 2: style.css

```
* {
  box-sizing: border-box;
}

body {
  margin: 0;
  min-height: 100vh;
  padding: 32px 16px;
  display: grid;
  place-items: center;
  background: linear-gradient(135deg, #eeeaff, #eaf7ff);
  color: #20243a;
  font-family: Arial, Helvetica, sans-serif;
}

.app {
  width: min(100%, 560px);
  padding: 34px;
  background: white;
  border-radius: 20px;
  box-shadow: 0 18px 55px rgba(45, 48, 90, 0.12);
}

.eyebrow {
  margin: 0 0 8px;
  color: #6558d3;
  font-size: 0.75rem;
  font-weight: bold;
  letter-spacing: 2px;
}

h1 {
  margin: 0;
  font-size: clamp(2rem, 7vw, 2.8rem);
}

.subtitle {
  margin: 10px 0 26px;
  color: #73798c;
}

form {
  display: flex;
  gap: 10px;
}

input {
  flex: 1;
  min-width: 0;
  padding: 13px 14px;
  border: 1px solid #dfe2ec;
  border-radius: 9px;
  font-size: 1rem;
}

input:focus {
  outline: 2px solid #bcb5ff;
  border-color: #6558d3;
}

button {
  border: 0;
  border-radius: 9px;
  padding: 12px 16px;
  background: #6558d3;
  color: white;
  font-weight: bold;
  cursor: pointer;
}

button:hover {
  background: #4f43b5;
}

.summary {
  display: flex;
  justify-content: space-between;
  align-items: center;
  gap: 12px;
  margin: 24px 0 12px;
  color: #73798c;
  font-size: 0.9rem;
}

.text-button {
  padding: 6px 0;
  color: #6558d3;
  background: transparent;
```

```
}

.text-button:hover {
  color: #4f43b5;
  background: transparent;
}

#task-list {
  list-style: none;
  margin: 0;
  padding: 0;
}

.task {
  display: flex;
  align-items: center;
  gap: 12px;
  padding: 14px 0;
  border-bottom: 1px solid #eff0f5;
}

.task input[type="checkbox"] {
  flex: 0 0 auto;
  width: 18px;
  height: 18px;
  accent-color: #6558d3;
}

.task-text {
  flex: 1;
  overflow-wrap: anywhere;
}

.task.completed .task-text {
  color: #9297a7;
  text-decoration: line-through;
}

.delete-button {
  padding: 7px 10px;
  background: #fff0f0;
  color: #b83232;
}

.delete-button:hover {
  background: #ffdddd;
}

#empty-message {
  padding: 22px 0;
  color: #9297a7;
  text-align: center;
}

@media (max-width: 430px) {
  .app { padding: 24px 18px; }
  form { flex-direction: column; }
  form button { width: 100%; }
}
```

File 3: script.js

```
const form = document.querySelector("#task-form");
const input = document.querySelector("#task-input");
const list = document.querySelector("#task-list");
const count = document.querySelector("#task-count");
const emptyMessage = document.querySelector("#empty-message");
const clearCompletedButton = document.querySelector("#clear-completed");

let tasks = [];

function renderTasks() {
  list.replaceChildren();

  tasks.forEach((task) => {
    const item = document.createElement("li");
    item.className = task.completed ? "task completed" : "task";

    const checkbox = document.createElement("input");
    checkbox.type = "checkbox";
    checkbox.checked = task.completed;
    checkbox.setAttribute("aria-label", "Mark task as completed");
    checkbox.addEventListener("change", () => {
      task.completed = checkbox.checked;
      renderTasks();
    });

    const text = document.createElement("span");
    text.className = "task-text";
    text.textContent = task.text;

    const deleteButton = document.createElement("button");
    deleteButton.className = "delete-button";
    deleteButton.type = "button";
    deleteButton.textContent = "Delete";
    deleteButton.addEventListener("click", () => {
      tasks = tasks.filter((entry) => entry.id !== task.id);
      renderTasks();
    });

    item.append(checkbox, text, deleteButton);
    list.append(item);
  });

  const remaining = tasks.filter((task) => !task.completed).length;
  count.textContent = `${remaining} task${remaining === 1 ? "" : "s"} left`;
  emptyMessage.hidden = tasks.length > 0;
}

form.addEventListener("submit", (event) => {
  event.preventDefault();
  const text = input.value.trim();

  if (!text) return;

  tasks.push({
    id: Date.now() + Math.random(),
    text,
    completed: false
  });

  input.value = "";
  input.focus();
  renderTasks();
});

clearCompletedButton.addEventListener("click", () => {
  tasks = tasks.filter((task) => !task.completed);
  renderTasks();
});

renderTasks();
```